

Financial Fraud Detection

Analysis Report

End-to-End Data Analytics Project

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Tools: Python | PostgreSQL | Tableau | Jupyter Notebook

1,296,675	7,506	0.58%
Total Transactions	Fraud Cases	Fraud Rate

01 Project Overview

This project performs an end-to-end financial fraud detection analysis on 1.29 million simulated credit card transactions spanning January 2019 to December 2020. Using Python, PostgreSQL, and Tableau, the analysis uncovers hidden fraud patterns across time, geography, transaction categories, and customer demographics — providing actionable business recommendations to reduce financial losses.

Problem Statement: Credit card fraud costs businesses billions of dollars annually. With fraud accounting for only 0.58% of all transactions, it is nearly impossible to detect using traditional methods. This project uses advanced data analytics to surface hidden fraud patterns and recommend targeted interventions.

02 Dataset & Tools

Attribute	Details
Source	Kaggle — Credit Card Transactions Fraud Detection (Kartik Shenoy)
Total Records	1,296,675 transactions
Time Period	January 2019 – December 2020
Customers	1,000 unique customers
Merchants	800 unique merchants
Fraud Cases	7,506 (0.58% of total)
Key Columns	amount, category, timestamp, location, gender, dob, is_fraud

Tool	Purpose
Python (Pandas, Matplotlib, Seaborn)	Data cleaning, EDA, feature engineering, visualizations
PostgreSQL (pgAdmin)	SQL queries and relational database analysis
Tableau Public	Interactive dashboard with 5 chart types
Jupyter Notebook	Analysis environment and documentation

03 Key Insights

01 Fraud Rate is Extremely Low

- Only 0.58% of all 1.29M transactions are fraudulent
- 7,506 fraud cases detected out of 1,296,675 total transactions
- This imbalance highlights the need for pattern-based fraud detection

02 High-Risk Transaction Categories

- Grocery POS leads with 1,743 fraud cases — highest by volume
- Shopping Net follows with 1,713 cases and \$1.71M total fraud amount
- Online categories (shopping_net + misc_net) account for 35% of all fraud
- Travel has the fewest cases but highest average fraud amount (\$1,051)

03 Fraud Transactions Are 7.8x Higher in Amount

- Average fraud transaction: \$531 vs \$67 for normal transactions
- Travel fraud averages \$1,051 — nearly 9x higher than normal travel transactions
- High transaction amount is a strong fraud signal across all categories

04 Peak Fraud Hours: 10PM to 3AM

- 51% of ALL fraud occurs between 10PM and 3AM
- Hour 22 (10PM) = 1,931 cases — single highest fraud hour
- Hour 23 (11PM) = 1,904 cases — second highest
- Daytime hours (4AM–11AM) show minimal fraud activity (40–60 cases/hour)

05 Geographic Fraud Hotspots

- New York (NY) leads with 555 fraud cases and \$295,548 in losses
- Texas (TX) second with 479 cases and \$265,806 in losses
- Pennsylvania (PA) third with 458 cases and \$244,624 in losses
- NY + TX + PA together account for over \$800,000 in fraud losses

06 Gender is Not a Strong Fraud Predictor

- Male fraud rate: 0.64% vs Female: 0.53% — very close
- Absolute cases: Males = 3,771, Females = 3,735
- Gender alone cannot reliably identify fraudulent transactions

04 SQL Analysis Summary

Six analytical SQL queries were executed in PostgreSQL (pgAdmin) to validate and extend findings from the Python EDA phase:

Query	Purpose	Key Finding
Q1 — Fraud Summary	Overall fraud vs normal comparison	Fraud avg \$531 vs \$67 normal (7.8x)
Q2 — Top Categories	Highest fraud by category	Grocery POS #1 (1,743), Shopping Net #2 (1,713)
Q3 — Gender Analysis	Fraud rate by gender	Male 0.64% vs Female 0.53%
Q4 — Top States	Geographic fraud distribution	NY > TX > PA in total fraud losses
Q5 — Peak Hours	Time-based fraud patterns	10PM (1,931) and 11PM (1,904) are peak hours
Q6 — High Value Fraud	Transactions above avg fraud amount	Shopping categories average \$1,300+ per fraud

05 Tableau Dashboard

An interactive Tableau dashboard was built using five different chart types to visually communicate all key fraud patterns to both technical and non-technical stakeholders:

Chart	Type	Insight Communicated
Fraud vs Normal %	Pie Chart	Only 0.58% of transactions are fraudulent
Fraud by Category	Treemap	Grocery & online shopping are highest risk
Fraud by State	Geographic Map	NY, TX, PA are geographic hotspots
Fraud by Hour	Heatmap	10PM–11PM are peak fraud hours
Fraud by Gender	Bar Chart	Gender is not a strong fraud predictor

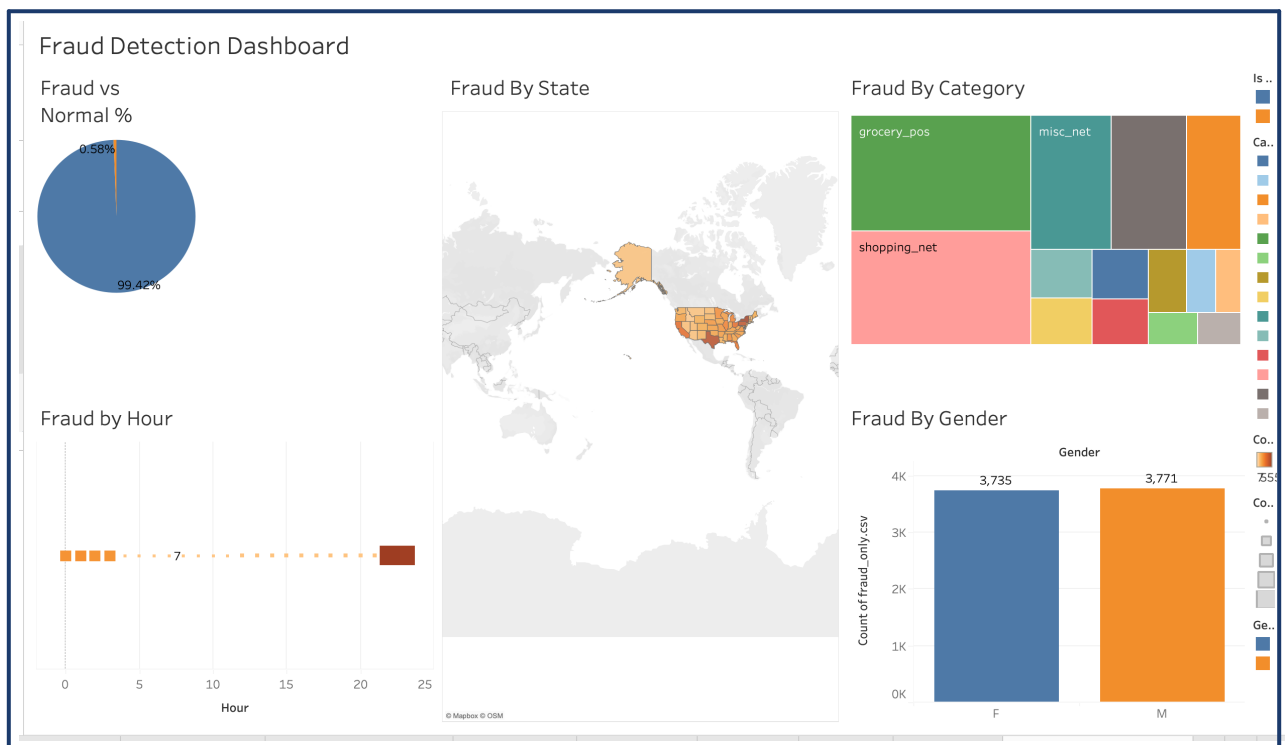


Figure 1: Financial Fraud Detection Dashboard — Tableau

06 Business Recommendations

"Implementing targeted fraud monitoring during 10PM–3AM and flagging high-value online transactions could reduce fraud losses by up to 51%"

01

Increase Monitoring Between 10PM–3AM

Deploy real-time fraud detection algorithms during late night hours. Over 51% of all fraud occurs in this 5-hour window.

02

Flag High-Value Transactions in Shopping Categories

Automatically flag transactions above \$500 in grocery_pos, shopping_net, and misc_net for secondary verification.

03

Implement Geo-Based Fraud Alerts

Set up automatic alerts for high-value transactions from NY, TX, and PA. These three states account for \$800,000+ in fraud losses.

04

Add Friction for Online Transactions

Introduce step-up authentication for shopping_net and misc_net. Online fraud in these categories totals \$2.4M — the largest fraud channel.

07 Project Workflow

Phase	Tool	Tasks Completed
Phase 1 — Data Loading	Python / Pandas	Loaded 1.29M rows, explored shape, columns, data types
Phase 2 — Data Cleaning	Python / Pandas	Converted dates, extracted hour & age features
Phase 3 — EDA	Python / Pandas	Answered 4 business questions using groupby, value_counts
Phase 4 — Visualization	Matplotlib / Seaborn	Built 4 charts: bar, boxplot, gender comparison
Phase 5 — SQL Analysis	PostgreSQL / pgAdmin	Ran 6 queries covering categories, states, hours
Phase 6 — Dashboard	Tableau	Built 5-chart interactive dashboard
Phase 7 — Documentation	GitHub / README	Structured repository, wrote project README and report

Radhika Chaturvedi | Data Analyst Portfolio
Project | 2024

Tools: Python | PostgreSQL | Tableau |
Jupyter